

PortPulse AI

Container Congestion & Port Operations Optimiser

AI Track

IBM BoB AI Innovation Hackathon 2026

Team Portvector

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The Port Congestion Problem

Why container terminals need better decision support



The Challenge

- Identify bottlenecks before they cascade
- Model interventions before committing resources
- Generate grounded, explainable recommendations
- Visualise the full operational picture at a glance

L1 Challenge (IBM BoB Hackathon):

Container Congestion Predictor &

Port Operations Optimiser

From Port Data to Operational Decisions

End-to-end pipeline: data → optimisation → decision support



Key Design Principle

- The AI Operations Copilot reads engine outputs only — it cannot invent, modify, or override a single calculated number.
- Every AI-stated fact traces directly to the optimizer or congestion engine — verified by a dedicated 100-check test suite.
- The Copilot is architected so a live LLM can replace the rule-based templates without touching any engine or the dashboard.
- No external API, no internet connection, no cloud account required — the entire system runs entirely locally.

How PortPulse AI Works

Component architecture — data flows and module responsibilities



Data: simulated 15-vessel / 5-berth dataset (72-hour planning horizon) — no live port feed

No external API, no cloud services, no API keys required — entirely local Python application

IBM Bob used as engineering assistant during development (implementation, testing, documentation)

Scenario Analysis: What Changes When Conditions Change?

20% service-time reduction — verified engine results

BASELINE		20% Service-Time Reduction ▼ RE-OPTIMIZE ▼ OR-Tools SCIP MIP re-runs on modified inputs	SCENARIO RESULT	
Scheduled Vessels	15 / 15		Scheduled Vessels	15 / 15
Total Waiting Time	39 h		Total Waiting Time	27 h
Avg Waiting Time	2.6 h		Avg Waiting Time	1.8 h
Max Waiting Time	16 h		Max Waiting Time	10 h
Waiting Vessels	6		Waiting Vessels	6
Demurrage	\$132,600		Demurrage	\$88,200
Berth Operating Cost	\$60,300		Berth Operating Cost	\$53,100
Total Modeled Cost	\$192,900		Total Modeled Cost	\$141,300
Bottleneck Berth	Berth 3		Bottleneck Berth	Berth 2

AI Copilot automatically detects and warns: bottleneck shifts from Berth 3 to Berth 2 after intervention

IBM Technology & Development Workflow

Accurate representation of IBM involvement in this project

IBM Bob (BoB AI Assistant)

Role in this project: Engineering Assistance

- Used as an AI coding and engineering assistant throughout development
- Assisted with: Python implementation, OR-Tools MIP formulation
- Assisted with: deterministic AI Copilot logic and test suite design
- Assisted with: documentation, architecture decisions, code review
- Not part of the runtime application — no IBM API called at run time

Runtime Technology Stack

Python 3.10+	All application logic
Streamlit	Dashboard framework
OR-Tools (SCIP MIP)	Berth scheduling optimizer
pandas / numpy	Data processing
Plotly	Interactive visualisations
CSV data files	Simulated port dataset

Modeled Operational Impact

20% service-time reduction scenario vs. baseline — verified engine values

39h → 27h

-12h

Total Waiting Time

\$192.9k → \$141.3k

-\$51.6k

Total Modeled Cost

Berth 3 → Berth 2

Bottleneck Shift

Bottleneck Shift

15 / 15

0 infeasible

Vessels Scheduled

\$132.6k → \$88.2k

-\$44.4k

Demurrage

\$60.3k → \$53.1k

-\$7.2k

Berth Operating Cost

Results are from our simulated/test dataset and represent modeled scenario outcomes, not real-world port performance. No savings are guaranteed.

Team Portvector

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